

Problem 1 (The Timelike Tourist): A tourist lives in a spacetime of events described by coordinates $\mathbf{P} = (t, x, y)$, where t is the time coordinate, and x and y are Cartesian spatial coordinates, of each event. There are one thousand *must-see events* that happen in this spacetime, labeled by $\mathbf{P}_k = (t_k, x_k, y_k)$ for $k \in \{1, 2, \dots, 1000\}$. Begin with

$$\mathbf{P}_1 = (t_1, x_1, y_1) = (0, 0, 0),$$

and for $k > 1$ define

$$t_k = (17 y_{k-1} + 23) \bmod 10007,$$

$$x_k = (31 t_{k-1} + 37) \bmod 10007,$$

$$y_k = (43 x_{k-1} + 47) \bmod 10007,$$

where “mod 10007” denotes the least nonnegative remainder.

Between any two events $\mathbf{P}_i$ and $\mathbf{P}_j$, define the spacetime interval

$$D(\mathbf{P}_i, \mathbf{P}_j) = D(\mathbf{P}_j, \mathbf{P}_i) = -100 \left(1 - \frac{2}{9 + x_i^2 + y_i^2} - \frac{2}{9 + x_j^2 + y_j^2} \right) (t_j - t_i)^2 + (x_j - x_i)^2 + (y_j - y_i)^2.$$

A tourist may travel directly from $\mathbf{P}_i$ to $\mathbf{P}_j$ if and only if the step is *chronological*; that is, if $i \rightarrow j$ moves forward in time, and the spacetime interval between the two events is negative¹:

$$t_i < t_j \quad \text{and} \quad D(\mathbf{P}_i, \mathbf{P}_j) < 0.$$

A *journey* is a sequence of distinct events in which every consecutive step is chronological. A journey may begin and end at any events of your choosing. **Whoever submits the journey containing the most must-see events wins.**

Give your submission as the labels of the events, listed in the order they are visited. For example, $(1, 2, 3, 4)$ represents the proposed journey

$$\mathbf{P}_1 \longrightarrow \mathbf{P}_2 \longrightarrow \mathbf{P}_3 \longrightarrow \mathbf{P}_4.$$

¹Physically, $D < 0$ means that the two events can be connected by travel at less than the speed of light.